

Astro4 – Extragalactic Astrophysics – Computer Project 2

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Learning Goals:

The goals of this project are for you:

- 1) to be able to use Python/jupyter notebook to work with spectral data and extend your python programming experience
- 2) to know what an initial mass function (IMF) is and why it is relevant.
- 3) to be able to characterize stellar and galaxy spectra and why different galaxy types have different spectra.
- 4) to understand and be able to explain the limitations of galaxy spectra and spectral modeling and the reasons for these limitations
- 5) to understand what simple stellar populations models are and their limitations.

Partners: You are expected to have one or two *discussion partners* with whom you discuss your strategy to solve this task and discuss the results and answers to the questions. You should work in groups of 2-3 people (no more). I strongly recommend that you each take turn in taking charge of the programming at the keyboard. You submit only one report for the group. Do not split the work between you. You all need to have an in depth understanding of all the issues of this work as this project is an important part of the exam curriculum for this course.

A few words on the report you are expected to submit.

Deadline: Your written assignment is due the latest on **Monday January 5, 2026 at 10:00 pm**. Remember that you are welcome to submit earlier too!. Submit your report as a PDF file on Absalon. It is a group submission, so only 1 report per group. Please do a careful piece of work you can take pride in. The report must be legible, neat, and organized to receive full credit. Use full grammatical sentences. The more well-organized report that is easy to read and understand will receive a higher grade. To pass this part of the class, your report must obtain a grade of 75%.

You are expected to do all the exercises in this project and to support your answers with your own jupyter notebooks (or Python programs if you choose this option) and figures generated from it. Further instructions are given in Part VI at the back of this document. Do check the start of this section to acquaint yourself with what results and figures you need to include in the report.

SPECTRA OF STARS AND GALAXIES

The light from galaxies comes from the stars that reside in them. The goal of this project is to determine the relative distribution of stars of different spectral type that reside in different galaxies. You will be using the range of stellar template spectra in your folder to reproduce galaxy spectra. The stellar templates are from a catalog of Jacoby et al. (1984, ApJS, 56, 257).

In the following, python commands will be written in `typewriter` font for easy identification. Use the material distributed to complete this project.

Part I: Read in the data – inspect the data

1. First, make a sub-folder in your top directory called ‘project2’: Copy from Absalon/Computer Exercise 2 the data file called Exercise2data.zip.
Unzip the file. The zipped file contains a set of stellar template spectra in ASCII format (i.e., text) and a set of galaxy spectra in FITS format.
2. Next, identify the spectra of the stars in your folder. The stellar spectra are named by their spectral type, for example: A2V.asc, G2V.asc, etc. Look for files of type ‘*V.asc’, where ‘*’ is a wildcard –

meaning ‘any entry’. Recall that the ‘V’ implies stars on the main sequence. The format are normal ascii text files with two columns and the units of the data are rest frame wavelengths ($\text{\AA}$) for the first column, and flux density ($\text{erg/s/cm}^2/\text{\AA}$) for the second column.

3. Write a small program to read in and plot each of the stellar spectra. Save each plot as a .jpeg or .pdf file. One option is to read an input list of files and loop over each entry, reading and plotting the file. For help with the syntax, check the plotting programmes in exercise 1.

4. **Do this at home:** Use the textbook and in particular Fig. 1.1 to compare the spectra of each of your stars with their spectral classification. Discuss what the spectral type means and take notes in preparation for your report. *You should finish this before the next computer-session.*

5. Let’s get accustomed with scaling and plotting spectra (spend only 10 – 15 mins on this):

i. Try scaling some of the spectra by multiplying the flux array by a number and plot it (or overplot it).

ii. Take the ratio of one stellar spectrum to another. Plot the result to the screen.

What is the ratio of? Flux density or wavelengths?

iii. Try plotting an early type spectrum and over-plotting a late-type spectrum in a different color. Make sure that both spectra are visible in the plot. What do you see? Surprised?

You will reuse these small Jupyter Notebooks/Python programmes in the following.

Part II: Spectral Composites

The stellar spectra are scaled to be at the same (random) distance from us such that they have the appropriate relative brightness as their spectral type reflects. You can make a model composite spectrum of groups of stars by adding stellar spectra. By altering the relative contribution (i.e. number) of the stars of different spectral type you alter the appearance of the model spectrum; that is, the continuum slope and strengths of different spectral features. We can generate model spectra of stellar clusters and galaxies by adding spectra of stars in the right proportion according to the stellar content of these objects.

Write a program to generate your composite models. Remember to write you own comments inside the program to explain what it does.

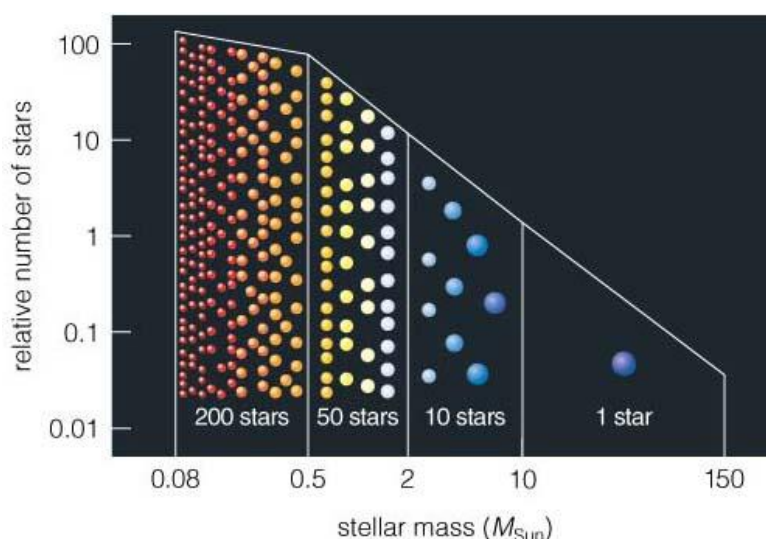


Figure 1: Initial Mass Function. Demographics of newborn stars. This schematic diagram shows how many stars of each mass are produced for every star greater than $10 M_{\odot}$ in an episode of star formation. Very massive stars are relatively rare, lower-mass stars are relatively common.

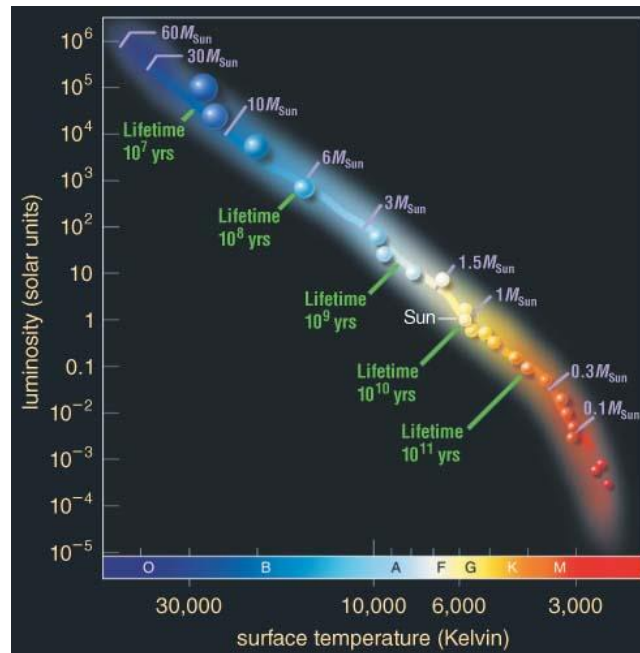


Figure 2: The approximate variation of masses and lifetimes along the main-sequence in the H-R diagram. Notice the stark difference between low and high mass stars.

A) Spectrum of an IMF

Here we will make a composite spectrum using the ‘Initial Mass Function’ by adding stellar spectra from your library *in the relative proportion* shown in Figure 1.

i) First, we need to decide which spectral types to place in each of the boxes. *To make your work easier later on, keep all stars of a particular stellar type in the same box. I suggest having O stars in one box, B stars in another, A,F,G stars in a 3rd, K stars in a 4th, and M stars in a 5th box.* The relationship between spectral types and stellar mass is given in Figure 2; *Alternatively you may use a table listing stellar type with masses.*

Make a simple approximation of the distribution like what happens in a star-forming event. That is, nature did not just make 1 stellar type in one of the five boxes or an equal proportion of stars of different stellar types in each box. Figure 1 illustrates with relative numbers that there are 200 stars with a mass range between 0.08 and 0.5 solar masses, 50 stars with masses between 0.5 and 2 solar masses etc..

Remember also that within each stellar class (OBAFGKM), sub-classes with numbers 0,1,2,...9, that reflect stars with different masses and temperatures too. The IMF then dictates that there are more stars with a higher number than with lower numbers.

ii) Write a code that sums the spectra of a chosen number of the stars in each box from Fig. 1 into their own spectrum such that you can plot the individual spectrum from each box. Give the spectrum from each box its own variable name (e.g., ‘box1’). Thereafter, you can sum all boxes to produce the spectrum of the final composite (i.e., `model=box1+box2+box3+box4+box5`). Call the final composite ‘composite’ or ‘model’. When you sum two spectra, what physical quantity do you actually sum (flux density or wavelength)?

iii) Compare the boxes by over-plotting their spectra with different colors onto the final composite. Plot the sum of the boxes starting from the box with the least massive stars and then add the boxes with the most massive stars one by one and discuss what you observe. In the end, compare the composite spectrum of all the stars to the spectrum of the most massive star you have among the stellar spectra. Explain what you see, and if this is what you expected?

B) For the report: Comment on this composite spectrum. What does it represent? How is it different in shape and appearance from individual stars in your folder?

Note: You can comment later as you write the report, but you should discuss this with your group members. Make sure to discuss similar questions later in this exercise with your group – and do ask questions if you need clarification.

Part III: Stellar composites in Galaxy data

The goal of this exercise is to determine the relative contribution of stellar types in galaxies. For that you will create composite spectra of stars and compare it with spectra of galaxies. Before you start each task, discuss within your group the best strategy to solve the task and why this makes sense. Take notes on this discussion, so you don't forget it later. *The approach and the reasoning behind it is a large part of this exercise. We suggest you make a new copy of the program you made for Part II, give it a new name and use it for this exercise.*

A) You have spectra of two galaxies in your folder named 'gal1.fits' and 'gal2.fits'. The flux density is on an arbitrary scale.

Write a small program to read in the data. Plot the spectra of each "mystery" galaxy. Keep in mind that you need to generate the wavelength array yourself based on the information in the image header. You may check the notebook `Fits_demo.ipynb` from Exercise 1 to understand how to read in FITS spectra in Python, and retrieving header information and data values. For example, a FITS header *may* contain:

```
CRVAL1 =      1545.  
CRPIX1 =         1.  
CDELTA1 =        3.  
CD1_1 =         3.
```

This means that the first pixel has index number 1 and value 1545Å. The wavelength increments are in 3Å – meaning that pixel number 2 has value 1545Å + 3Å = 1548Å. *What values do your spectra have?*

B) Start with galaxy 1 (gal1.fits). Compare the composite model you made in Part II with the galaxy spectrum by plotting the two. Discuss: how do the amplitudes of the stellar IMF composite spectral model compare with the galaxy spectra? Explain what you see. Why is there an amplitude difference?

C) Change the relative scaling of the individual boxes (which then changes the number of stars of a particular spectral type in your composite) until you are able to reproduce the spectrum of galaxy1. Use your new code to add, scale, and compare the spectra. Note that a relative scaling is sufficient and the scaling factors need not be integers.

Hint: Don't just randomly change the relative number of spectral type or you'll never finish. Be systematic and think about what makes sense. Follow the steps below.

i) Taking the ratio of the spectrum that you try to reproduce and your model spectrum is very useful for discovering where your model may be insufficient or finding the approximate scaling factor to apply in the end. *To generate the ratio of two spectra, the X-axis of both spectra must be the same (Python works with the indices, not the wavelength values).* Are they the same?

If not, you need to interpolate the flux with the wavelength array being the same, e.g. with `scipy.interpolate.griddata`, and import the Python module:

```
import scipy.interpolate.
```

This linear interpolation program projects your original spectrum (your pair of wavelength and flux-density arrays) onto another pair of arrays of a different length and step-size. Given a new wavelength

array you'll get the new flux levels at each index in the wavelength array. Check how to use the correct syntax for this function. The syntax is normally:

```
new_flux=scipy.interpolate(old_wavelength,old_flux, new_wavelength)
```

ii) First, aim at getting the overall shape of your model spectrum similar to the galaxy spectrum by using a different scaling factor of each box that goes into your model. That is, do *not* change the distribution of stellar types within the boxes at this time. Once you have the overall shape correct, you can start refining the relative scaling of the stars *within* each box for a closer match. This way you converge much faster. The final step should be to scale the total sum of the boxes with a single factor (representing to some extent the distance to the galaxy).

iii) Once you are close to matching the target spectrum, **stop and decide what constitutes a good match and why?** Clearly, the flux levels must match – but how well? Ignore the emission lines; you cannot match them with stellar spectra. Use the ratio of data/model to estimate the percentage level (see hint below). Note down your strategy. You are likely to revise it later, but make sure you note down all the steps you took, and what made you reconsider your approach or your match criteria. *[This is part of the scientific method and what professional astronomers do all the time. No-one will believe you if you just quote a number without any explanation of how you got it. And if your steps cannot be reproduced, it is not a valid scientific method.]*

Hint: The ratio plot shows to which %-level your match is good. Overplot horizontal lines at 1.0 and at 0.9 and 1.1. Then you can see if your model is good to within 10%. This way, it is easier to see and quantify how well your model matches. Aim for ~10% or better.

C) Once you believe you are within your match criteria, make sure to plot the galaxy spectrum with the model on top using a different colour. Also, plot a ratio of the two spectra for your report (match the X-axis). Describe your results and discuss what have you learned from this task.

D) Determine the limitations of the model: the maximum and minimum number of stars (bluest and reddest) that provide an equally good match to the galaxy spectrum as you found in IIIC above.

Here you will modify the numbers of the *bluest* stars in your model from III.C in a systematic way to see how unique your solution is. By the *bluest stars*, it means those stellar types that provide the bluest light for this particular galaxy – it may be an ‘A’ star if there are no O and B stars. We suggest you chose (the sum of) two stellar types to be the ‘bluest’ stars (but no more), especially if the bluest stellar type in your model has a very small and negligible contribution to the spectral range in the blue. You are looking for the highest and lowest number of these ‘blue stars’ that give a similarly good match to within your match criteria. Note that if you chose a very small confidence limit (i.e., match criteria < 5%), this may be hard to accomplish.

Hint I: Define a wavelength range to focus on first. Then determine the approximate continuum level in that region. Thereafter, see how many stars you can add or subtract until that continuum level hits the extreme 10% limits in your ratio plot. **Hint II:** For each of the extreme number of blue stars, *work on a copy of the code*, that gave the best matching galaxy model. Then you always have a record of everything you did for the report.

When you find the highest and lowest number of ‘blue stars’ that produces a similarly good match, make pdf or jpg figures of these case studies to support your findings.

For the report you need to **derive the highest and lowest number** of the ‘bluest stars’: the number (of min. or max. blue stars) relative to the total number of stars in your best model spectrum, i.e., a fractional change: Compute the number of ‘blue’ stars that you can change (up and down, respectively) from your best model, relative to the total number of stars that you have in your best model.

Provide a table in the report with the number of blue stars (i) in your best model, (ii) in your blue-min model, (iii) in your blue-max model and (iv) the total number of stars in your best model. Then compute the ratio $(\text{blue-best} - \text{blue-min})/\text{total}(\text{best-model})$ and $(\text{blue-max} - \text{blue-best})/\text{total}(\text{best-model})$. It is instructive to compute the ratios in both numbers and in percent (%).

E) Repeat ‘D’ for the number of ‘reddest’ stars (i.e., K and M stars). Be sure to include the statistics in a table form.

F) For the report, you need to comment on what III ‘D’ and ‘E’ has taught you and what you conclude based on this exercise. Discuss this with your group either now or later.

G) Repeat steps A) – C) for gal2.fits. *Skip steps D, E and F for gal2.fits.*

H) Compare the results from your work on galaxy1 with that of galaxy2. Do you have any (qualified) guess at what type of galaxy each of them is? Explain your reasoning in the report.

Part IV: Use of Simple Stellar Populations

As you know, not all stars in our Galaxy are main-sequence stars (which you have in your template library) and the (spectral) appearance of stars changes as they move off the main-sequence. Astronomers have made more sophisticated models of stellar populations than you have created to try to determine the stellar composition of galaxies.

One such model is ‘simple stellar populations’ which are composite stellar populations models based on observed spectra of stars put together to reflect the age of a group of stars that were born at essentially the same time. It is similar to determine the composite spectrum of a (huge) star cluster at different stages in its life. Astronomers can use this to establish if a galaxy has any population of young stars and what age they may be.

First, investigate the content of the subfolder ‘ssp’.

Here you will find spectra of stellar populations of different ages as labeled (‘Gy’ means Giga year (10^9 yr), ‘My’ means Mega year (10^6 yr), and ‘ky’ means kilo year (10^3 yr). Open the text files with an editor to check that the wavelength scale is the same. In that case, reading and adding the data is easier. First plot the spectra so you can see what they look like. Pay attention to the characteristics of each age-group.

A) See if you can find a *single* stellar population spectrum among the ‘simple stellar populations (SSP)’, which most closely matches that of the first composite you made in section II.A (the IMF model spectrum, based on Figure 1). Which is it? Does it make sense and why (not)? Overplot your IMF model spectrum on each of the SSP models to identify the best SSP model and show that figure in the report.

B) Using these stellar population spectra of different ages, use the technique you developed in Part III of combining individual spectra to reproduce the spectrum of galaxy1 and to find the *approximate* age(s) of the stellar populations in galaxy1. Note that the SSP-spectra are arbitrarily scaled by the same factor. *A general match is sufficient.* If you want to compute an average age, compute a weighted average using the scaling factors that you use to build the stellar population of galaxy1. Again, make sure you support your findings with explanations of your strategy and add plots, tables, and diagrams.

C) Repeat “B)” for galaxy2.

D) Comment on what you find. How do the galaxies differ? How are they similar? Does this make sense based on what you found in Part III? Does this part of the project give you a better sense of how to classify the galaxies?

Part V: Chemical Abundances.

Go to folder 'ssp2' :

In this directory you will find another 2 sets of simple stellar populations. As the filenames indicate one set is based on stellar population models with approximately solar metallicity ($Z=0.02$), the other set is based on similar models with super-solar metallicity ($Z=0.05$). The SSP models in Part IV have solar metallicity, $Z=0.02$ (and are the same as those in this folder). The wavelengths are identical for all the SSP models.

- A)** Write a small program to plot these SSP models – comparing for each age the solar and super-solar models. **Scale the SSP models to approximately match in the red part of the spectrum.** What is typical for both of these models? *Hint:* the wavelengths are identical for all the SSP models.
- B)** For the set of super-solar SSP models, find the best **single** SSP model to galaxy 2 (gal2.fits) from part III and part IV. You will likely need to scale the SSP model as before. Explain what potentially is meant by an 'age-metallicity degeneracy'?

Part VI: Important report items

You are expected to do all the exercises on this assignment and to support your answers with your own Python programmes and figures generated from it.

Your report should include:

- A.** The python notebooks/codes you wrote to solve the exercises. You can attach them as separate electronic files on Absalon e.g as a ZIP file.
- B.** Support your work with plots and diagrams that show what you have found. Make sure all figures and diagrams have figure number, axes, tick-marks, and are properly labeled with appropriate numbers and axes titles and that all these are legible (i.e., readable). Figures, labels and axis numbers should be large enough to be read well.
- C.** Mark or highlight features or parts of the diagram or figure that you refer to in your description and which are important (such that there is absolutely no doubt about what you refer to).
- D.** Be sure to number and label your figures clearly and sequentially, as they are relevant, i.e. Fig. 10 should come after and not before Fig. 9.
- E.** Place your figures and tables in the text where you talk about it in the text. If you need to discuss a figure or table shown already, refer to the relevant figure/table. Only the spectra of the stars from Part I, you can place in the Appendix (owing to the large number of plots).
- F.** Make sure you answer the questions in the order they are posed (see below).
- G.** Please explain any assumptions you make and the reason(s) why you make them.
- H.** Also, be careful to explain your approach and the scientific reasons why you adopt it.

The items listed below have more details. If you are unsure what is meant by the text in items 1. – 13., then ask!

For your written report, you should answer the following questions (be careful to answer all the questions for each item listed), and in this order:

Part II

1. Show the plots of each stellar spectral type in the Appendix. **Briefly**, comment on the main spectral features and characteristics, which reveal the spectral type of each star in your stellar library. What signifies the spectral type of an early type star compared to a late type one? For example, why does the spectrum peak at shorter wavelengths for early type stars while late type stars have the opposite slope? Why are the spectral features different for different stars? What creates these absorption lines?

2a. Show the spectral composite from Part II.A. List the contribution of stellar types that you made for each box with the number of stars of each spectral type that you used to generate this spectrum. What do you see in this spectrum? How does it compare to the spectrum of an individual star?

2b. Show the diagrams you made in II.A: Plot on the same figure the spectra of each of the 5 boxes, and the final sum of all the boxes containing the full IMF. Comment on what you see.

2c. Show a diagram where you over-plot the composite on the spectrum of an O-star and describe what you see and conclude.

2d. What exactly does this composite represent? (Hint: consider what went into making this composite spectrum. Which types of stars and their (approximate) age are they – where would they fall in the H-R diagram?). How is it different in shape, appearance and amplitude from individual stars and the galaxy spectra. Why do we call it an IMF?

Part III

3. First describe what a visual inspection of the two galaxy spectra tells you about the contents of the galaxies without any computer analysis.

4. Explain the flux amplitude difference between the galaxy spectra to the stellar spectra. What can you learn from comparing the spectral composite from part II to the two galaxy spectra ?

5. Describe your strategy for solving Part III – see the first paragraph of “Part III”. Explain why you chose this scientific approach. Describe your criteria for a good match between the galaxy spectrum and your model. Explain how you came to decide on this. Comment on how useful and appropriate this is and explain why your approach makes sense scientifically.

6a. What were the steps you took to solve this task of reproducing galaxy1 with individual stellar spectra in Part III step C? What is your final best match? Support your findings with figures as explained in the exercises: Show the model over-plotted on the data, and also ratio plot(s). Include a table (or similar) of how many stars of each stellar type you have in each box and in total for your best galaxy model. *(If you prefer to show your scaling factors, include also the total number of stars for each type and in total).*

6b. Report on your findings to Part III steps D) – E). Show all your numbers (e.g. in a table. Please do **not** mix the results in with the text – make your results easy to check). List the statistics in both numbers and percentage; see the exercise description). List the numbers of stars that you measure and use for your calculations and to get your statistics. Comment on the statistics you computed.

6c. How unique is your model? If the range of acceptable models is large or even modest then a particular solution or model fit is not very unique and our understanding of that object will be limited because we cannot accurately say what its stellar composition is. Comment on whether your model is considered unique. Consider this for both the blue and the red stars..

6d. Comment on what Part III ‘D’ and ‘E’ has taught you and what you can conclude based on this exercise.

7. Repeat item 6a for galaxy2. Report on the results for this galaxy.

8a. Include your findings from Part III, H) to give your best suggestion to the classification of the two galaxies you have worked on. Explain your reasoning.

8b. Explain the (positive) narrow “spikes” seen in one of the galaxy spectra. What are their actual names and what is their origin?

9. First list what type of information you know about a star if you know its specific spectral type, e.g., F4V. Check the H-R diagram in Figure 2 for inspiration. Given your model with the number distribution of each spectral type that makes up the spectrum of the galaxy, can you identify what other property of the galaxy we can measure using this model? It is a basic, but important property of the galaxy as a whole.

10. Part IV

Explain briefly what a Simple Stellar Population (SSP) is.

11. Part IV

Based on the work you did in Part IV, briefly describe your approach for determining the approximate ages of the stellar populations for the two galaxies. Is it the same you used for Part III or how does it differ?

12. Part IV

a) Describe your results of Part IV A) – C) and explain why your match is good. Include in a table the composition of the SSP models that you found to be a good match to your galaxies. Use the table and your scaling of each SSP to compute the weighted average age of the stellar populations.

b) Consider this average age and the SSP age spread you have determined: can these stellar populations tell you what might have occurred in the past? How do the two galaxies compare and what have you learned about them?

c) Provide an updated guess at the nature and classification of these two galaxies. Look at the age-spread that you see of the stellar populations. Does that motivate you to update your conclusion about any of the galaxies?

13. Part V.

a) Describe your results from Part V steps A and B. What do you generally see when you compare for a given age the SSP models based on different metallicities? What is characteristic of these models? What is the physical explanation for the differences?

b) Which single SSP model fits best to galaxy2? Make a direct comparison with the most dominant stellar population age that you determined in Part IV (questions 11-12).

c) What is potentially meant by ‘age-metallicity degeneracy’?